

**The Financial Opportunity Cost of Continued Navy Service
for Primary-Care Physicians: A General-Internist Case**

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Purpose and Economic Framework

The U.S. Navy Medical Corps must understand the alternatives physicians face when deciding whether to remain after their initial obligation. Using a board-certified general internist in San Diego, this paper asks: How much financial compensation does a physician forgo by remaining in the Navy for four more years rather than accepting a feasible civilian position? General internal medicine serves as the primary-care case because it is a core Navy primary-care specialty with a published local civilian benchmark; family medicine and other primary-care fields pay differently, so the results should not be generalized to them without separate estimates. The U.S. Government Accountability Office (GAO, 2020) found that maximum military cash compensation trailed civilian medians in 21 of 27 medical and dental specialties reviewed and recommended better wage, bonus, and replacement-cost data. Those findings motivate this case study but do not prove a current Navy primary-care pay gap.

The question also has a macroeconomic dimension. Military physician pay is set administratively through statute and Department of Defense pay tables, while civilian pay adjusts to national and regional supply and demand for physicians. When civilian primary-care wages rise faster than administered military pay, the retention gap widens without any change in Navy policy. Location magnifies this: Marit Health (2026) reports that the San Diego internist average exceeds the California average by about 27% and the national average by about 40%, so the Navy competes for physicians in some of its highest-wage labor markets.

Opportunity cost is the value of the best forgone alternative: here, the civilian compensation stream a physician gives up by staying. The analysis reports the net cash cost of staying, defined as the present value of that civilian stream minus Navy cash compensation over the same period. Prior medical-school support is sunk; future pay, obligations, retirement accrual, and working conditions remain relevant (Frank et al., 2022). Mission, leadership, teaching, operational medicine, assignment control, workload, and family stability may also create compensating wage differentials. Therefore, the cash gap is not a complete valuation of either career and does not identify the bonus required for retention.

Method

The base case is an O-4 general internist just past 10 active-service years, with dependents, assigned to ZIP code 92134. The horizon is 2027-2030 in constant 2026 dollars. Rank remains O-4; basic pay advances from the over-10 to the over-12 rate. The analysis uses a 5% real discount rate and year-end payments. The civilian comparator is Marit Health's published San Diego internist all-employer average of \$459,057 in total compensation (average base of \$445,914 plus average bonus of \$13,142), held constant in real terms (Marit Health, 2026). Marit

builds its averages from self-reported, NPI-verified submissions after removing outliers and applying de-biasing adjustments. The figure is a mean rather than a median, so a few high earners can pull it upward, and it rests on a small number of local submissions. Its academic and non-academic averages (\$250,000 and \$471,727) also show wide dispersion by employer type. Because this is a mixed-employer, self-reported benchmark rather than a matched offer, the result remains conditional on its comparability.

Compensation Inputs

Appendix A, Table 1 itemizes Navy cash pay and the conditional retention bonus. Each annual gap is discounted as $(\text{civilian cash} - \text{Navy cash}) / (1.05)^t$, where $t = 1, 2, 3, 4$. The results therefore compare gross cash only; taxes, health coverage, retirement contributions, and other benefits are addressed under Direction of Bias below.

Career-Stage Horizon and Pension

The four-year window is not the decision a physician at 10 years of service actually faces. The realistic alternatives are separating now or remaining to the 20-year active-duty retirement threshold, and the base case ends at 14 years, the point at which staying looks least valuable because no pension has vested. The pension also needs its own discount rate. Blended Retirement System (BRS) retired pay is adjusted annually for inflation and backed by the federal government, so its risk resembles an inflation-protected Treasury security more than uncertain civilian earnings (Schofer, 2016). Treasury real yields on September 28, 2026, were 3.14% at 20 years and 3.28% at 30 years (U.S. Department of the Treasury, 2026), so the pension is discounted at 3% real while cash pay remains at 5%. For illustration, assuming a \$120,000 High-3 basic-pay base, 30 constant real payments, and a 3% pension rate, a \$48,000 BRS pension has a conditional present value of about \$700,000 at 10 years, \$887,000 at 18 years, and \$913,000 at 19 years (Department of Defense, n.d.); at 5%, the same values would be about \$453,000, \$669,000, and \$703,000. A physician who separates at 10 years forfeits that value entirely. These figures cannot simply be subtracted from the four-year cash gap because the horizons differ, but they show that the four-year gap overstates the net cost of staying for anyone who expects to reach 20 years. A complete comparison therefore sets the pension against the full remaining cash gap to 20 years.

Figure 2 makes that comparison at every service year from 10 to 19. At each year, it sets the present value of the pension preserved by staying to 20 against the present value of the civilian cash still to be forgone before 20, holding Navy gross cash at \$282,585 and civilian pay at \$459,057 in every remaining year. At 10 years, the ten-year remaining cash gap has a present value of about \$1,363,000, against a pension worth about \$700,000, so staying to 20 carries a net

cash cost of about \$663,000 (Appendix A, Table 2). Each additional year of service removes one year of forgone civilian pay from the remaining gap and brings the pension one year closer; because the same pension is discounted over fewer years, its present value rises even though the benefit itself does not change. The two lines cross at about 14.7 years, after which staying to 20 dominates on cash terms; by 19 years, the net value of staying is about \$745,000. Discounting the pension at 5% instead would move the break-even to about 16.1 years and raise the net cost at 10 years to about \$910,000, so the discount-rate choice is the single most influential pension assumption.

This pattern is the option value of staying: the pension exerts little pull at mid-career and a strong pull near eligibility. For retention policy, it implies that cash incentives matter most between roughly 10 and 15 years of service, before the pension begins to hold physicians in place, and matter much less afterward. Promotion to O-5 would raise both cash pay and the pension: an O-5 High-3 of about \$139,000 (the average of 2026 O-5 basic pay at 16 and 18 years of service) yields a \$55,700 BRS pension and, with cash pay unchanged, cuts the net cost of staying at 10 years to about \$550,000. Promotion and the tax effects described under Direction of Bias would each shift the break-even point earlier, while the chance of separating before 20 for other reasons would lower the pension's expected value and shift it later. Figure 2 also counts cash only; mission, deployments, assignment control, and family stability lie outside it and can push an individual decision in either direction.

Results

Figure 1 shows the annual comparison and cumulative gap. The base case produces an undiscounted cash gap in constant 2026 dollars of \$717,126 and a present-value gap of approximately \$636,209. At 3%, the present-value gap is \$666,715; at 7%, it is \$607,907. Without the assumed \$48,000 retention agreement, the four-year present-value gap rises by approximately \$170,206. These estimates compare compensation streams; they do not predict separation. Dahle (2021) illustrates substantial within-specialty pay variation; his examples do not establish the distribution of San Diego internist offers. Illustrative annual civilian salaries of \$350,000 and \$400,000 reduce the four-year present-value gap to about \$249,500 and \$426,800, respectively (undiscounted: about \$280,900 and \$480,900). These are sensitivity assumptions, not verified offers. A smaller matched gap makes other benefits more consequential; a larger gap strengthens the case for compensation changes.

Direction of Bias

Most of the items the model omits make the gross-cash gap larger than an after-tax, total-compensation comparison would show. First, BAH and BAS, \$64,926 per year in this case, are

excluded from federal income tax (Internal Revenue Service, 2025); at a 22% to 24% marginal rate, the exclusion is worth roughly \$14,000 to \$16,000 per year. Second, a servicemember keeps a legal residence for state tax purposes when assigned elsewhere (50 U.S.C. § 4001), so a Navy physician domiciled in a state without an income tax owes no California tax, whereas a civilian internist in San Diego would. At the benchmark salary, California tax for a married couple filing jointly is roughly \$34,000 per year before credits (California Franchise Tax Board, 2025). Third, holding rank at O-4 through 14 years understates Navy pay, because many physicians are promoted to O-5 in that window. Fourth, TRICARE family coverage during service and after retirement, BRS matching contributions of up to 5% of basic pay, and BRS continuation pay are excluded; retiree health coverage from retirement to Medicare eligibility can be worth thousands of dollars a year but varies by family, so it is left to individual estimates.

Some omissions run the other way. Civilian employers often pay malpractice premiums, contribute to retirement plans, and offer signing bonuses, none of which are in the civilian figure. On balance, the tax and pension effects appear large enough that the after-tax gap is probably materially smaller than the gross-cash estimate, although this paper does not quantify the net adjustment.

Policy Implications

The Navy should test a limited communication pilot alongside, not in place of, bonus advocacy. The case for communication rests on a specific, testable claim: mid-career physicians may underweight the conditional pension, tax-free allowances, and state-tax treatment when comparing offers, because civilian recruiters quote gross salary. If so, accurate total-compensation information is a low-cost intervention. If not, communication will not move retention, and pay or working conditions must carry the burden. Changing authorized bonus rates requires a formal pay-policy and funding process, so a communication effort can start sooner through existing channels. Its value would be correcting misunderstandings about real, accessible opportunities and existing compensation, not substituting advertising for inadequate pay.

A proposed design would enroll physicians whose obligations end within 12 months and randomly assign commands to enhanced communication or standard information, balancing specialty and years of service. Measure awareness at baseline and six months, agreement acceptance over 12 months, and actual retention through 24 months, accounting for communication spillover. Because only a limited number of internists reach the end of an obligation in any 12-month window, the pilot should pool specialties or run across several annual cohorts; a single-specialty, single-year trial would likely be underpowered to detect a meaningful retention effect. If awareness improves without a meaningful retention gain at acceptable cost,

stop expansion and prioritize compensation or working conditions. Inconclusive results require more evidence, not a claim of success.

Additional Options

Team-based staffing could shift appropriate work toward nurse practitioners and physician assistants while preserving physician capacity for complex care and readiness. VA evidence supports feasibility in selected populations but not one-for-one Navy substitution (Liu et al., 2020; Morgan et al., 2019). The Navy should also address workload, administrative burden, leadership support, deployment, and work-life concerns associated with burnout and turnover-related outcomes (Vie et al., 2024; Wilk et al., 2023).

Limitations

The model uses one rank, location, and civilian benchmark, and that benchmark is a self-reported mean drawn from a small local sample. It assumes bonus eligibility and does not quantify promotion, taxes, health coverage, retirement contributions, malpractice costs, moonlighting constraints, or probability-adjusted pension value; the Direction of Bias section describes their likely effect only qualitatively. It does not estimate Navy attrition, replacement costs, retention elasticity, or readiness losses. The ten-year comparison in Figure 2 holds Navy pay and rank constant, treats reaching 20 years as certain, and depends heavily on the pension discount rate, which is set from market real yields on a single date. Future analysis should add promotion, after-tax pay, and the probability of reaching retirement eligibility, and should compare matched career paths by specialty and service stage.

Conclusion

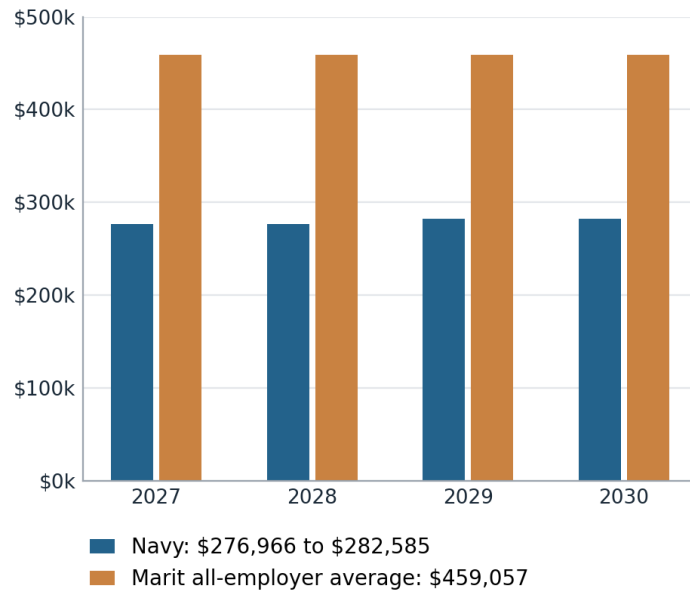
For the modeled O-4 internist, the four-year present-value gross-cash gap is approximately \$636,209 at 5%. That figure likely overstates the real cost of staying: it excludes tax advantages and promotion, and it ends before the pension that a physician at 10 years of service preserves by remaining to 20. Measured over the ten years to retirement eligibility, and valuing the inflation-protected pension at a 3% real rate, staying to 20 still carries a net gross-cash cost of about \$663,000 at 10 years of service, but that cost shrinks each year and reverses at about 14.7 years as the pension approaches. Even after tax adjustments, the gap is probably substantial, particularly in high-wage markets such as San Diego. The Navy should test targeted total-compensation communication alongside bonus advocacy, evaluate bonuses by marginal retention effect, improve working conditions, and collect better career-stage data.

Figure 1. Annual compensation and cumulative discounted cash gap

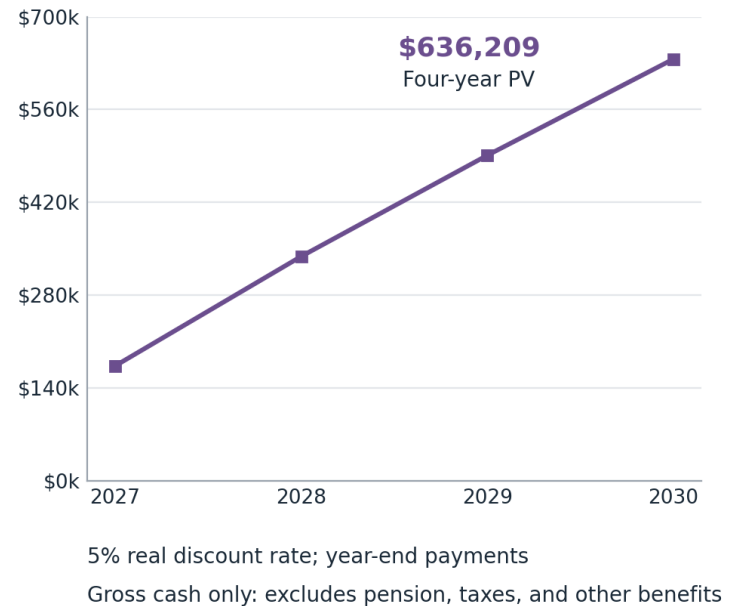
Internal medicine in San Diego

Illustrative Navy O-4 versus published local benchmark | Constant 2026 dollars

A Annual gross cash



B Cumulative discounted cash gap



Navy: O-4, over-10 years of service, with dependents, ZIP 92134; conditional \$48,000 annual retention bonus.

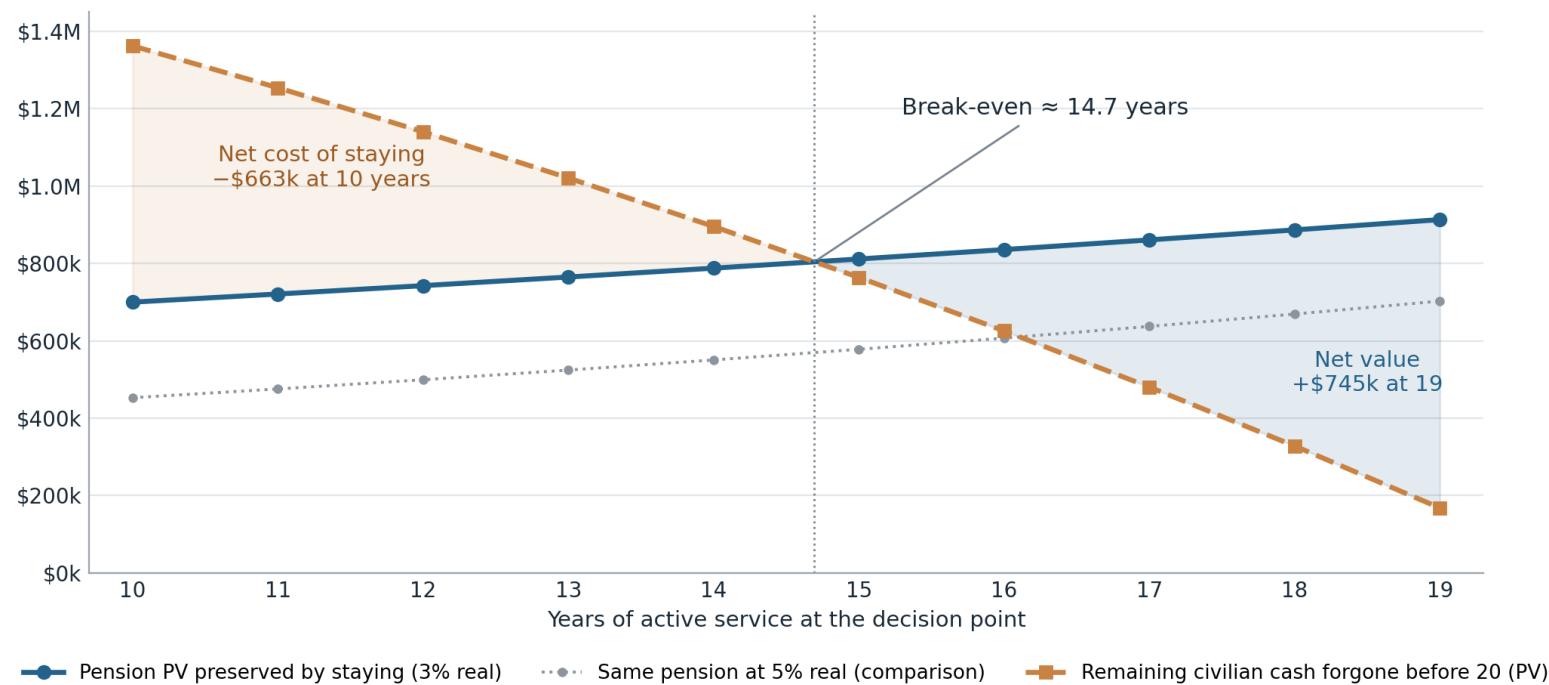
Sources: DFAS (2026a, 2026b, 2026c); Defense Travel Management Office BAH lookup; Marit Health (April 2026). Not a matched civilian offer.

Illustrative O-4 scenario; 5% real discounting. The civilian series is not a matched offer. Pension, taxes, and other benefit differences are excluded; see Direction of Bias.

Figure 2. Pension value preserved versus remaining cash gap, by years of service

Why staying gets easier: the pension pull

Present value at each year of service of staying to 20 | Constant 2026 dollars; pension at 3% real, cash at 5% real



Pension: \$48,000 BRS annuity (2% × 20 × \$120,000 High-3), 30 real payments, conditional on reaching 20; 3% ≈ Treasury real yields (Sept. 28, 2026).
Cash gap: \$459,057 civilian benchmark — \$282,585 Navy gross cash per year. Gross cash only. At 5% pension rate, break-even ≈ 16.1 years.

Each point is the present value at that year of service of staying to 20. The pension is discounted at 3% real and cash pay at 5% real; the dotted line shows the same pension at 5%. Illustrative and conditional on reaching 20 years; gross cash only. Values appear in Appendix A, Table 2.

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Appendix A. Compensation inputs

Navy component	2027–2028	2029–2030
Basic pay	\$113,040	\$118,660
BAH with dependents	\$60,984	\$60,984
BAS	\$3,942	\$3,942
Incentive pay	\$43,000	\$43,000
Board-certification pay	\$8,000	\$8,000
Assumed four-year retention bonus	\$48,000	\$48,000
Total Navy gross cash	\$276,966	\$282,585

Table 1. Base-case cash compensation. The retention bonus is conditional on eligibility and an executed agreement. BAH and BAS (\$64,926 per year) are excluded from federal income tax. Totals use unrounded inputs (Defense Finance and Accounting Service [DFAS], 2026a, 2026b, 2026c; Defense Travel Management Office, n.d.).

Table 2. Ten-year comparison: staying to 20 years, by years of service at the decision point

Years of service	Pension PV (3%)	Remaining cash gap PV (5%)	Net value of staying	Net if pension at 5%
10	\$700,059	\$1,362,667	–\$662,608	–\$909,675
11	\$721,061	\$1,254,329	–\$533,268	–\$778,687
12	\$742,693	\$1,140,574	–\$397,881	–\$641,149
13	\$764,974	\$1,021,131	–\$256,157	–\$496,735
14	\$787,923	\$895,716	–\$107,793	–\$345,100
15	\$811,561	\$764,030	\$47,531	–\$185,884
16	\$835,907	\$625,760	\$210,148	–\$18,706
17	\$860,985	\$480,576	\$380,409	\$156,830
18	\$886,814	\$328,133	\$558,681	\$341,144
19	\$913,419	\$168,068	\$745,350	\$534,672

Pension PV: \$48,000 BRS annuity ($2\% \times 20 \text{ years} \times \$120,000 \text{ High-3}$), 30 constant real payments beginning after year 20, discounted at 3% real to the decision point. Remaining cash gap PV: ($\$459,057 - \$282,585$) per remaining year to 20, year-end payments, discounted at 5% real. Net value = pension PV – remaining cash gap PV; last column: pension at 5%. Gross cash; conditional on reaching 20 years (Department of Defense, n.d.; Marit Health, 2026; U.S. Department of the Treasury, 2026).